Use work-energy to solve motion

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose work-energy when force and displacement are known.

SUCCESS CRITERIA - I can...

1. List all forces doing work.
2. Calculate the kinetic-energy change.
3. Check whether the result is physically possible.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A force parallel to motion does 15 J over 3 m. Find its size. What work does the same force do with zero displacement?

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $W_{net} = \frac{1}{2}m(v_f^2 - v_i^2)$.
Constant mass. Use the displacement along each force component; stopping removes the initial kinetic energy.

Worked reasoning: A 2 kg cart starts from rest. Constant net force 4 N acts through 3 m, doing 12 J. Then $\frac{1}{2}mv^2=12$ gives $v=\sqrt{12}\approx3.46$ m/s.

Your turn: Sketch a force diagram and an energy bar diagram for this motion. Identify where time is unnecessary.

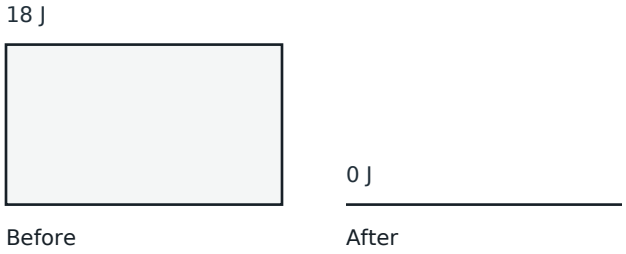
Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and interpret the model

A cart has 18 J kinetic energy and then stops. How much net work is done? Show an energy bar before and after.



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: ① ② ③

04 Apply the physics

A 2 kg lab cart moving at 6 m/s meets a constant 3 N resistive force. Complete the energy account and find its stopping distance. Predict the new stopping distance if its initial speed doubles; verify using kinetic energy.

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A calculation gives negative final kinetic energy after a long braking distance. What should you check?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Initial momentum is 0.4 kg·m/s and measured final momentum is 0.38 kg·m/s. Find percent deviation. Explain why this is not itself a measured uncertainty.
Use before and after snapshots to organize the momentum account.

Before	After

Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

07 Exit check

A 5 kg cart starts at rest. A 12 N pull and 2 N resistance act over 4 m. Find final speed. Use the energy model to show the quantities you compare; label units and any unknowns.

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 4 kg cart at 5 m/s slows to 1 m/s over 6 m. Find the constant net force component along its motion. Use the energy model to show the quantities you compare; label units and any unknowns.

Initial	Transfers	Final

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3